

Impact of 6G Network Performance on Manufacturing Efficiency in Smart Factories: A Machine Learning-Based Analytical Framework

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Abstract— Smart factories increasingly depend on connected machines, sensors, industrial communication networks and intelligent automation systems. As future industrial networks move toward 6G technologies, understanding the relationship between communication performance and manufacturing efficiency becomes an important research problem. This work presents a machine-learning-based analytical framework for studying network performance and manufacturing efficiency in a smart-factory environment. The proposed workflow includes data preparation, exploratory data analysis, feature engineering, network KPI analysis, manufacturing KPI analysis, relationship analysis and supervised machine learning. Logistic Regression, Decision Tree and Random Forest models are considered for efficiency-status prediction. The implementation also generates model results, predictions, prediction errors and analytical findings as project artifacts. The study is data-driven and observational; therefore, statistical relationships are interpreted as associations rather than causal effects. The proposed framework provides a foundation for future real-time industrial monitoring, edge intelligence and digital-twin applications.

Keywords— 6G, Smart Factory, Industrial IoT, Manufacturing Efficiency, Machine Learning, Network KPI, Predictive Analytics, Industry 4.0

I. INTRODUCTION

The rapid development of smart manufacturing has increased the dependence of industrial systems on reliable communication between machines, sensors, robots, controllers and cloud or edge platforms. Industrial automation applications generate continuous data and may require communication with high reliability, low latency and sufficient capacity.

Sixth-generation wireless communication, commonly referred to as 6G, is expected to extend the capabilities of future industrial networks through improved connectivity, intelligent network management, sensing capabilities and support for demanding applications. At the same time, manufacturing efficiency depends on several operational characteristics such as production performance, quality, utilization and process stability.

This work investigates these two domains using a structured smart-factory dataset and a machine-learning-based analytical workflow. Network performance indicators are analysed together with manufacturing indicators to identify measurable relationships and evaluate whether machine-learning models can predict an efficiency-status target.



Fig. 1. Overall analytical framework for connecting network performance with manufacturing efficiency.

II. RELATED WORK

Several studies have investigated low-power, reliable and high-performance communication systems for industrial automation. Recent research on 6G wireless communication for industrial automation has discussed future industrial scenarios, communication requirements, reliability, latency and challenges associated with highly connected manufacturing systems.

Machine learning has also been applied to industrial monitoring, prediction, anomaly detection and optimization. Data-driven methods can identify patterns in complex industrial datasets that may not be easily captured using fixed analytical rules. However, the relationship between network-level indicators and manufacturing-level outcomes depends on the characteristics and quality of the available data.

Based on these observations, the present work combines network KPI analysis, manufacturing KPI analysis and supervised machine learning within a single reproducible project workflow.

III. METHODOLOGY

The proposed work follows a structured data-analysis and machine-learning methodology. The complete project includes data preparation, exploratory analysis, network analysis, manufacturing analysis, relationship analysis and predictive modelling.

A. Dataset and Project Specifications

The project uses a structured smart-factory dataset containing variables representing network performance and manufacturing behaviour. The dataset is prepared for statistical analysis and supervised machine learning. The efficiency-status variable is used as the prediction target.

B. Data Preprocessing

The preprocessing stage includes inspection of the dataset, identification of relevant variables, preparation of data types and selection of suitable features for subsequent analysis. Data preparation is performed before model training to maintain a consistent modelling workflow.

C. Exploratory Data Analysis

Exploratory data analysis is performed to understand variable distributions, patterns and relationships. Descriptive statistics and visual analysis are used to identify useful characteristics of the dataset before applying machine-learning algorithms.

D. Feature Engineering

Relevant network and manufacturing variables are selected and prepared as model inputs. Feature engineering is performed according to the structure of the available dataset so that the resulting features can be used by the classification models.

E. Machine Learning Pipeline

The machine-learning workflow follows the sequence: Dataset -> Preprocessing -> Feature Selection -> Train/Test Split -> Model Training -> Prediction -> Evaluation -> Error Analysis.



Fig. 2. Machine-learning workflow used for efficiency-status prediction.

IV. NETWORK PERFORMANCE ANALYSIS

The network analysis stage examines communication-related indicators contained in the smart-factory dataset. Network KPIs provide information about the communication conditions associated with industrial operation.

The generated network findings and relationship files provide a structured representation of the observed network characteristics. These outputs can be extended with additional 6G indicators when real industrial measurements become available.

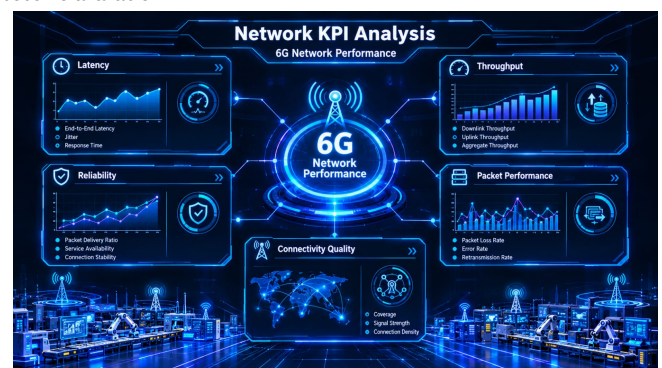


Fig. 3. Network KPI analysis and visualization.

V. MANUFACTURING EFFICIENCY ANALYSIS

Manufacturing-related variables are analysed to understand the operational performance represented in the dataset. These variables form the basis for interpreting the efficiency-status target used in the predictive analysis.

The manufacturing analysis is treated as a multidimensional problem. Individual indicators may describe different aspects of manufacturing performance, and their interpretation depends on the definitions used in the dataset.



Fig. 4. Manufacturing efficiency analysis.

VI. NETWORK–MANUFACTURING RELATIONSHIP ANALYSIS

The relationship-analysis stage examines statistical associations between network performance indicators and manufacturing indicators. Correlation and relationship outputs are stored as separate project artifacts.

These relationships should not be interpreted as proof of causation. An observed association may result from common underlying factors, dataset construction or variables that are not represented in the analysis. Controlled experiments and real industrial measurements are required to establish causal relationships.

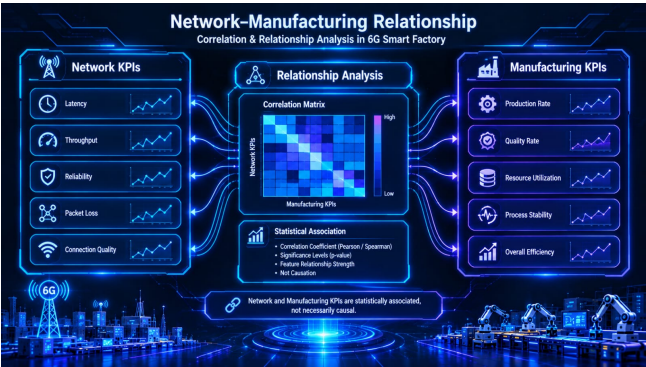


Fig. 5. Relationship analysis between network and manufacturing indicators.

VII. MACHINE LEARNING MODELS

A. Logistic Regression

Logistic Regression is considered as a baseline classification model. It provides a comparatively simple approach for estimating the probability of the efficiency-status classes from the selected input features.

B. Decision Tree

The Decision Tree model represents classification decisions through a hierarchical set of feature-based conditions. Its structure can provide an interpretable representation of the classification process.

C. Random Forest

Random Forest combines multiple decision trees to obtain an ensemble classifier. The method can represent nonlinear relationships and complex feature interactions present in the dataset.

The three models are trained and evaluated using the project modelling pipeline. Predictions and prediction errors are retained for further inspection.

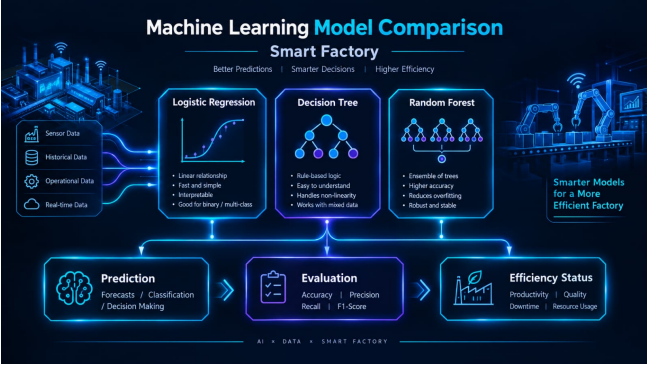


Fig. 6. Machine-learning model comparison and prediction workflow.

VIII. RESULTS AND DISCUSSION

The project generates dedicated result files containing model results, predictions and prediction errors. Additional outputs include network findings, manufacturing findings and network–manufacturing relationship summaries.

The numerical performance values are intentionally not fabricated in this paper. Exact accuracy, precision, recall, F1-score or other metrics should be reported directly from the generated project result files after verification of the final experiment.

The analytical workflow demonstrates how communication indicators and manufacturing indicators can be studied together and subsequently supplied to machine-learning models. Predictive performance, however, should be interpreted together with target balance, data quality and the intended industrial application.

IX. LIMITATIONS

The present study is based on a structured dataset rather than live measurements from a physical 6G smart-factory deployment. Therefore, the observed relationships describe the analysed data and should not automatically be generalized to every industrial environment.

Other limitations include dependence on feature selection, target definition, dataset representativeness and the evaluation protocol. Real industrial systems may additionally contain missing telemetry, equipment-specific behaviour, changing network conditions and concept drift.

X. FUTURE SCOPE

The framework can be extended toward real-time industrial IoT monitoring using sensors, edge devices and wireless communication infrastructure. Future development can include online anomaly detection, time-series forecasting, explainable machine learning and adaptive resource allocation.

Integration with a digital twin can further enable simulation of network and manufacturing conditions before operational decisions are applied. Hardware-assisted experiments can also provide real measurements for stronger validation of the relationships identified by the current study.

XI. CONCLUSION

This paper presented a machine-learning-based analytical framework for studying 6G network performance and manufacturing efficiency in smart factories. The proposed workflow integrates data preparation, exploratory analysis, network KPI analysis, manufacturing analysis, relationship analysis and supervised machine learning.

The framework provides a reproducible foundation for connecting future industrial wireless-network measurements with manufacturing intelligence. Although the current study is observational and dataset-dependent, the approach can be extended toward real-time monitoring, edge intelligence and digital-twin-based industrial decision support.

TABLE I. ANALYTICAL COMPONENTS

Component	Purpose
Network KPI analysis	Analyse communication-performance indicators.
Manufacturing analysis	Study manufacturing and efficiency indicators.
Relationship analysis	Identify statistical associations.
Classification	Predict efficiency-status classes.
Error analysis	Inspect prediction errors.

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Project Repository: github.com/snehassneha4578-collab/Project-Impact-of-6-G-Network-Performance-on-Manufacturing-Efficiency-in-Smart-Factories